

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Rough work is to be done in the last page of main supplementary, please don't write anything on the question paper.
5. Indicate clearly, the option(s) you attempt along with its respective question no.
6. Figures to the right indicate marks.

SECTION-I

Q-1 Answer the following questions.

- a) What is the role of knowledge in solving AI problems? Up to what extent and level of knowledge is required for it? 3
- b) Give two examples of problem having following characteristics: 3
 1. Decomposable
 2. Recoverable
 3. Universal predictability
- c) What are the possible reasons for using heuristics? Differentiate between heuristic search and heuristic function. 3
- d) Write down the advantages of BFS over DFS. 2

Q-2

- [A] What is the Turing test? Does AI aim at human-level intelligence? Are computers the right kind of machine to be made intelligent? 4
- [B] What are the characteristics of the problem that are to be analyzed when choosing an appropriate method to solve the problem? Explain. 4
- [C] Analyze 8-puzzle problem with respect to the seven problem characteristics. 4

OR

Q-2

- [A] Draw the semantic network for the following sentence. 4
'Every dog in the town has bitten the constable'
- [B] What are the conditions to be satisfied to find an optimal path to a goal, if any path to a goal exists? 4
- [C] Define an appropriate state space representation for the game of Chess. 4

Q-3

- [A] Differentiate between cut and fail with the help of an example. 4
- [B] Explain: Learning in ANN. What is meant by generalization in feed forward network? 4
- [C] What are the basic knowledge representation issues? What are the considerations in knowledge representation? What is the role of knowledge acquisition in inference engine? 4

OR

Q-3

- [A] Write a prolog code to define clauses for relations: grandparents, parents and children. Prove, if somebody is a father of father of x, he will be a grandfather of x. 4

- [B] Distinguish between Declarative and procedural knowledge. Give any two examples for the same. 4
- [C] Convert the following into FOPL. 4
1. All employees earning more than 60,000 Rs. /year, pay income tax.
 2. No employee of University earns more than the VC.
 3. European vegetarians do not take milk, but Indian vegetarians do.
 4. Anything anyone eats and isn't killed by is fruit.

SECTION-II

- Q-4
- a) Define AI from your perspective. The goal of AI should be to build machines that help people in their intellectual tasks rather than to do those tasks. Justify this statement. 3
 - b) What is the need for prototype construction phase in the development cycle of an Expert system? 4
 - c) What is the use of resolution? Explain with an example? 4

- Q-5
- [A] Trace the operation of A* algorithm by taking a suitable example. 4
 - [B] What are the potentialities and limitations of Neural Networks? 4
 - [C] How to invent heuristic function for any problem? Explain in brief. 4

OR

- Q-5
- [A] How the frames are organized? What are the advantages and disadvantages of frames and semantic nets? Why the scripts are required? Can it be considered as a variant of frames? 4
 - [B] What is ALPHA cutoff and BETA cutoff? Explain benefits of it in searching. Also give one example for the same. 4
 - [C] Write a prolog program to find factorial of a given number using recursion. 4

- Q-6
- [A] What is the main drawback of back-propagation network? What is the reason of it? Whether momentum term is required for learning in a back-propagation network? What is the effect of this term? 4
 - [B] Explain the key idea of Bayesian statistics for representing knowledge. How Bayesian Belief Nets are used to classify items for a given problem? 4

OR

- [B] Trace the execution of the constraint satisfaction procedure in solving the crypt arithmetic problem: 4

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- [C] What are the basic characteristics of ES? What are the difficulties in developing an Expert System? Describe the main applications of Expert Systems. 4

OR

- [C] Derive the parse tree for the sentence using and writing the appropriate rules: 4

"Reserve Bank raises interest rates 0.5% in order to control the inflation".